Divisor of Algebraic Curves

MA4091 Seminar Matematika I

Gilbert Allister (10123021)
Dosen Pembimbing: Rizal Afgani, Ph.D.
Dosen Penguji 1: idk
Dosen Penguji 2: idk

Program Studi Sarjana Matematika
Fakultas Matematika dan Ilmu Pengetahuan Alam
Institut Teknologi Bandung

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Motivation

- Let k be an algebraically closed field. A polynomial $f \in k[x]$ is determined, up to a constant, by its zeros and their multiplicities:

$$f = c\,(x - a_1)^{m_1} \cdots (x - a_r)^{m_r}.$$

- In the same way, a rational function $f = g/h$ on the projective line $\mathbb{P}^1 = k \cup \{\infty\}$ is determined, up to a constant, by its **zeros and poles**. We record them in a formal sum, the **divisor** of f.

Example 1

For $f = \dfrac{(x-1)^2}{x}$ we get $\operatorname{div}(f) = 2\,[1] - [0] - [\infty]$: a double zero at 1 and simple poles at 0 and ∞.

- **Question.** On an algebraic curve X, fix points P_i and integers n_i. How large is the space of rational functions with poles of order at most n_i at P_i and no other poles? Divisors are the language for this question; the Riemann–Roch theorem is the answer.

Motivation

- **Generalizing zeros and poles:** divisors formalize the study of zeros and poles of rational (meromorphic) functions on an algebraic curve.
- **Transition to schemes:** scheme theory extends classical varieties over an algebraically closed field, so that arithmetic situations (arbitrary base rings) and non-reduced structures can be handled in one language.
- **Geometric invariants:** Weil and Cartier divisors are used to construct line bundles, which lead to central results such as the **Riemann–Roch theorem**.

Aim of my Undergraduate Thesis

The aims of my undergraduate thesis are:

1. To understand the basic objects of algebraic geometry.
2. To understand the basics of scheme theory and divisors of algebraic curves.
3. To identify the role of scheme theory in proving the Grothendieck–Riemann–Roch (GRR) theorem and its corollary, the Riemann–Roch (RR) theorem for curves.

This seminar covers the first two aims: the definitions and the main statements, from affine varieties up to divisors on a curve.

Outline of this section

1. Variety
2. Zariski Topology
3. Hilbert's Nullstellensatz
4. Affine Scheme

Variety

Definition 2 (Algebraic set)

Let k be an algebraically closed field and $A = k[X_1, \ldots, X_n]$. For an ideal J of A and a subset $X \subseteq k^n$, define

$$V(J) = \{a \in k^n \mid f(a) = 0 \ \forall f \in J\}, \qquad I(X) = \{f \in A \mid f(a) = 0 \ \forall a \in X\}.$$

$V(J)$ is the **algebraic set** defined by J; $I(X)$ is the **ideal of** X.

Definition 3 (Affine variety)

A non-empty algebraic set X is **irreducible** if it is not the union of two proper algebraic subsets. An **affine variety** is an irreducible algebraic set; its **coordinate ring** is $k[X] = A/I(X)$.

Example. In k^2: $V(Y - X^2)$ is an affine variety (a parabola); $V(XY) = V(X) \cup V(Y)$ is not irreducible (two lines).

Zariski Topology

Theorem 4

Let I, J and J_α ($\alpha \in \Lambda$) be ideals of $A = k[X_1, \ldots, X_n]$. Then

1. $V(\mathbf{0}) = k^n$ *and* $V(A) = \emptyset$;
2. *if* $I \subseteq J$, *then* $V(J) \subseteq V(I)$;
3. $\bigcap_{\alpha\in\Lambda} V(J_\alpha) = V\left(\sum_{\alpha\in\Lambda} J_\alpha\right)$;
4. $V(I) \cup V(J) = V(IJ) = V(I \cap J)$.

- By (1), (3) and (4), the algebraic sets are the closed sets of a topology on k^n.

Definition 5 (Zariski topology)

The **Zariski topology** on k^n is the topology whose closed sets are exactly the algebraic sets $V(J)$.

- For $n = 1$ the closed sets are k itself and the finite subsets of k, so the topology is not Hausdorff.

Radical Ideal

- Before stating the Nullstellensatz, we define the radical of an ideal in order to simplify the statement.

Definition 6 (Radical ideal)

Let I be an ideal of $A = k[X_1, \ldots, X_n]$. The **radical** of I is

$$\operatorname{rad} I = \{f \in A : f^m \in I \text{ for some integer } m \geq 1\}.$$

If $I = \operatorname{rad} I$, we say that I is a **radical ideal**.

- $\operatorname{rad} I$ is an ideal containing I, and $V(\operatorname{rad} I) = V(I)$.
- Every prime ideal is radical, and $I(X)$ is a radical ideal for every $X \subseteq k^n$.

Example 7

In $k[X]$ we have $\operatorname{rad}(X^2) = (X)$. Both ideals define the same algebraic set $\{0\}$, so the algebraic set does not remember the multiplicity.

Hilbert's Nullstellensatz

Theorem 8 (Nullstellensatz, weak form)

Let k be an algebraically closed field and $A = k[X_1, \ldots, X_n]$. If $\mathfrak{m}$ is a maximal ideal of A, then

$$\mathfrak{m} = (X_1 - a_1,\ X_2 - a_2,\ \ldots,\ X_n - a_n)$$

for some point $(a_1, a_2, \ldots, a_n) \in k^n$.

Theorem 9 (Nullstellensatz, strong form)

Let k be an algebraically closed field and $A = k[X_1, \ldots, X_n]$. If J is an ideal of A with $J \neq A$, then $V(J) \neq \emptyset$. Furthermore, for every ideal J of A,

$$I(V(J)) = \operatorname{rad} J.$$

The Algebra–Geometry Dictionary

Corollary 10

The maps $X \mapsto I(X)$ and $J \mapsto V(J)$ are mutually inverse, inclusion-reversing bijections:

Geometry (in k^n)		**Algebra** (in $A = k[X_1, \ldots, X_n]$)
algebraic sets	$\longleftrightarrow$	radical ideals
affine varieties	$\longleftrightarrow$	prime ideals
points	$\longleftrightarrow$	maximal ideals

- So the geometry of k^n can be recovered from the ring A alone.
- **Idea of scheme theory:** replace A by an arbitrary commutative ring with 1, and take *all* prime ideals as the points of the space.

Affine Scheme

Definition 11 (Spectrum of a ring)

Let A be a commutative ring with 1. The **spectrum** of A is the set $\operatorname{Spec} A = \{\mathfrak{p} \subset A \mid \mathfrak{p} \text{ is a prime ideal}\}$. For an ideal $\mathfrak{a}$ of A and an element $f \in A$, define

$$V(\mathfrak{a}) = \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \supseteq \mathfrak{a}\}, \qquad D(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}.$$

- The sets $V(\mathfrak{a})$ satisfy the same four properties as before, so they are the closed sets of the **Zariski topology** on $\operatorname{Spec} A$. The sets $D(f)$ form a basis of open sets.
- An **affine scheme** is $\operatorname{Spec} A$ together with its structure sheaf $\mathcal{O}_{\operatorname{Spec} A}$, the "functions" on $\operatorname{Spec} A$ (made precise in the next section).

Example 12

$\operatorname{Spec} k[X] = \{(X - a) \mid a \in k\} \cup \{(0)\}$. The points $(X - a)$ are closed; (0) is the **generic point**, whose closure is the whole space.

Outline of this section

1. Presheaf
2. Sheaf
3. Morphism of Sheaves
4. Sheafification
5. Schemes
6. Morphism of Schemes

Presheaf

Definition 13 (Presheaf)

Let X be a topological space. A **presheaf** $\mathcal{F}$ of abelian groups on X consists of

- an abelian group $\mathcal{F}(U)$ for every open set $U \subseteq X$, and
- a group homomorphism $\rho_{UV} \colon \mathcal{F}(U) \to \mathcal{F}(V)$ for every inclusion $V \subseteq U$ of open sets,

such that $\mathcal{F}(\emptyset) = 0$, $\rho_{UU} = \mathrm{id}_{\mathcal{F}(U)}$, and $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$ whenever $W \subseteq V \subseteq U$.

- Elements $s \in \mathcal{F}(U)$ are called **sections** of $\mathcal{F}$ over U; we write $s|_V = \rho_{UV}(s)$ and call ρ_{UV} the **restriction map**. Presheaves of rings are defined in the same way.
- The **stalk** of $\mathcal{F}$ at a point $p \in X$ is

$$\mathcal{F}_p = \varinjlim_{U \ni p} \mathcal{F}(U) = \{(U, s) \mid p \in U,\ s \in \mathcal{F}(U)\}/\sim,$$

where $(U, s) \sim (V, t)$ if $s|_W = t|_W$ for some open W with $p \in W \subseteq U \cap V$.

Sheaf

Definition 14 (Sheaf)

A presheaf $\mathcal{F}$ on X is a **sheaf** if for every open set $U \subseteq X$ and every open cover $\{V_i\}_i$ of U:

1. (**Locality**) if $s \in \mathcal{F}(U)$ satisfies $s|_{V_i} = 0$ for all i, then $s = 0$;
2. (**Gluing**) if $s_i \in \mathcal{F}(V_i)$ satisfy $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$ for all i, j, then there exists $s \in \mathcal{F}(U)$ with $s|_{V_i} = s_i$ for all i.

By locality, the section s in the gluing axiom is unique.

Example 15

- On any topological space X, $\mathcal{F}(U) = \{\text{continuous functions } U \to \mathbb{R}\}$ with the usual restriction of functions is a sheaf.
- For $X = \mathbb{R}$, $\mathcal{F}(U) = \{\text{bounded continuous functions } U \to \mathbb{R}\}$ is a presheaf but not a sheaf: the functions $s_i(x) = x$ on $V_i = (-i, i)$, $i \geq 1$, are bounded and agree on overlaps, but they glue to an unbounded function on $\mathbb{R}$.

Morphism of Sheaves

Definition 16 (Morphism of (pre)sheaves)

Let $\mathcal{F}, \mathcal{G}$ be presheaves on X. A **morphism** $\varphi : \mathcal{F} \to \mathcal{G}$ is a collection of group homomorphisms $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$, one for each open set $U \subseteq X$, such that for every inclusion $V \subseteq U$ the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \rho_{UV}} & & \downarrow{\scriptstyle \rho'_{UV}} \\ \mathcal{F}(V) & \xrightarrow[\varphi(V)]{} & \mathcal{G}(V) \end{array}$$

A morphism of sheaves is a morphism of the underlying presheaves.

Theorem 17

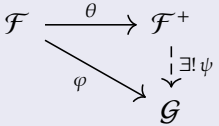
A morphism φ induces homomorphisms on stalks $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$. If $\mathcal{F}$ and $\mathcal{G}$ are sheaves, then φ is an isomorphism if and only if φ_p is an isomorphism for every $p \in X$.

Sheafification

- **Problem.** For a morphism of sheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$, the presheaves $U \mapsto \operatorname{im} \varphi(U)$ and $U \mapsto \operatorname{coker} \varphi(U)$ are in general not sheaves.

Theorem 18 (Sheafification)

Let $\mathcal{F}$ be a presheaf on X. There exist a sheaf $\mathcal{F}^+$ and a morphism $\theta\colon \mathcal{F} \to \mathcal{F}^+$ such that for every sheaf $\mathcal{G}$ and every morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$ there is a unique morphism $\psi\colon \mathcal{F}^+ \to \mathcal{G}$ with $\varphi = \psi \circ \theta$:



The pair $(\mathcal{F}^+, \theta)$ is unique up to unique isomorphism.

- $\mathcal{F}^+$ is called the **sheaf associated to** $\mathcal{F}$. It has the same stalks: $\mathcal{F}_p \cong \mathcal{F}_p^+$ for all $p \in X$.
- If $\mathcal{F}$ is already a sheaf, then θ is an isomorphism.

Schemes

Definition 19 (Locally ringed space)

A **ringed space** is a pair $(X, \mathcal{O}_X)$ of a topological space X and a sheaf of rings $\mathcal{O}_X$ on X. It is a **locally ringed space** if every stalk $\mathcal{O}_{X,p}$ is a local ring.

Theorem 20 (Structure sheaf of Spec A)

Let A be a ring. There is a sheaf of rings $\mathcal{O}$ on Spec A *such that*

$$\mathcal{O}(D(f)) \cong A_f \text{ for all } f \in A, \qquad \mathcal{O}_{\mathfrak{p}} \cong A_{\mathfrak{p}} \text{ for all } \mathfrak{p} \in \operatorname{Spec} A.$$

In particular $\mathcal{O}(\operatorname{Spec} A) \cong A$, *and* $(\operatorname{Spec} A, \mathcal{O})$ *is a locally ringed space.*

Definition 21 (Scheme)

An **affine scheme** is a locally ringed space isomorphic to $(\operatorname{Spec} A, \mathcal{O})$ for some ring A. A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ in which every point has an open neighbourhood U such that $(U, \mathcal{O}_X|_U)$ is an affine scheme.

Examples of Schemes

Example 22 (Affine space)

$\mathbb{A}^n_k = \operatorname{Spec} k[X_1, \ldots, X_n]$. By the Nullstellensatz, its closed points are the points of k^n; its other points are the prime ideals that are not maximal, i.e. the generic points of the subvarieties of positive dimension.

Example 23 (Projective line)

Take two copies of the affine line, $U_0 = \operatorname{Spec} k[x]$ and $U_1 = \operatorname{Spec} k[y]$, and glue the open sets $D(x) \subset U_0$ and $D(y) \subset U_1$ along the isomorphism given by

$$k[y, y^{-1}] \xrightarrow{\sim} k[x, x^{-1}], \qquad y \mapsto x^{-1}.$$

The result is a scheme $\mathbb{P}^1_k$. It is not affine: $\mathcal{O}(\mathbb{P}^1_k) = k$, while $\mathbb{P}^1_k$ has more than one point.

- This is the scheme-theoretic version of $\mathbb{P}^1 = k \cup \{\infty\}$ from the motivation: U_0 contains the finite points and U_1 contains ∞ (the point $y = 0$).

Morphism of Schemes

Definition 24 (Morphism of schemes)

A **morphism of schemes** $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$, where

- $f\colon X \to Y$ is a continuous map, and
- $f^\#\colon \mathcal{O}_Y \to f_*\mathcal{O}_X$ is a morphism of sheaves of rings on Y, where $(f_*\mathcal{O}_X)(V) = \mathcal{O}_X(f^{-1}(V))$,

such that for every $p \in X$ the induced map on stalks $f_p^\#\colon \mathcal{O}_{Y,f(p)} \to \mathcal{O}_{X,p}$ is a **local homomorphism**: $(f_p^\#)^{-1}(\mathfrak{m}_{X,p}) = \mathfrak{m}_{Y,f(p)}$.

Theorem 25

Every ring homomorphism $\varphi\colon A \to B$ induces a morphism of schemes $\operatorname{Spec} B \to \operatorname{Spec} A$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$, *and every morphism of schemes* $\operatorname{Spec} B \to \operatorname{Spec} A$ *arises from a unique φ:*

$$\operatorname{Hom}_{Schemes}(\operatorname{Spec} B, \operatorname{Spec} A) \cong \operatorname{Hom}_{Rings}(A, B).$$

A **scheme over** k is a scheme X together with a morphism $X \to \operatorname{Spec} k$.

Outline of this section

1 Algebraic Curves
2 Weil Divisor
3 Principal Divisors and the Class Group
4 Cartier Divisor
5 Line Bundles and the Riemann–Roch Theorem

Algebraic Curves

Definition 26 (Curve)

Let k be an algebraically closed field. A **curve** over k is an integral scheme X of dimension 1, proper over k, all of whose local rings are regular.

- In classical language, X is a nonsingular projective curve, for example $\mathbb{P}^1_k$.
- X has a unique generic point ξ. The local ring $K(X) = \mathcal{O}_{X,\xi}$ is a field, the **function field** of X; its elements are the rational functions on X.
- For every closed point $P \in X$, the local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring with fraction field $K(X)$. Its valuation

$$v_P \colon K(X)^* \to \mathbb{Z}$$

is the order of vanishing at P: f has a **zero** of order $v_P(f)$ if $v_P(f) > 0$, and a **pole** of order $-v_P(f)$ if $v_P(f) < 0$.

Example 27

For $X = \mathbb{P}^1_k$: $K(X) = k(x)$, and $v_\infty(g/h) = \deg h - \deg g$ for polynomials $g, h \neq 0$.

Weil Divisor

Let X be a curve over k.

Definition 28 (Weil divisor)

A **Weil divisor** on X is a formal finite sum

$$D = \sum_P n_P P, \qquad n_P \in \mathbb{Z},$$

over the closed points P of X, where $n_P = 0$ for all but finitely many P. The Weil divisors form the free abelian group $\mathrm{Div}(X)$ generated by the closed points of X.

- The **degree** of D is $\deg D = \sum_P n_P \in \mathbb{Z}$. This gives a group homomorphism $\deg\colon \mathrm{Div}(X) \to \mathbb{Z}$.
- D is **effective**, written $D \geq 0$, if $n_P \geq 0$ for all P.

Example 29

On $\mathbb{P}^1_k$: $D = 2[1] - [0] - [\infty]$ is a Weil divisor of degree 0; it is not effective.

Principal Divisors and the Class Group

Definition 30 (Principal divisor)

The **divisor of** a rational function $f \in K(X)^*$ is $\operatorname{div}(f) = \sum_P v_P(f)\, P$; this is a finite sum, since f has only finitely many zeros and poles. A divisor of this form is called **principal**.

Since $v_P(fg) = v_P(f) + v_P(g)$, the map $\operatorname{div}\colon K(X)^* \to \operatorname{Div}(X)$ is a group homomorphism.

Definition 31 (Linear equivalence, class group)

Two divisors D, D' are **linearly equivalent**, written $D \sim D'$, if $D - D' = \operatorname{div}(f)$ for some $f \in K(X)^*$. The **divisor class group** is $\operatorname{Cl}(X) = \operatorname{Div}(X)/\{\text{principal divisors}\}$.

Theorem 32

On a curve X, every principal divisor has degree 0: a rational function has as many zeros as poles, counted with multiplicity. Hence deg *induces a homomorphism* $\operatorname{Cl}(X) \to \mathbb{Z}$. *For* $X = \mathbb{P}^1_k$ *it is an isomorphism:* $\operatorname{Cl}(\mathbb{P}^1_k) \cong \mathbb{Z}$.

Cartier Divisor

- Let $\mathcal{K}$ be the constant sheaf $K(X)$ on X, $\mathcal{K}^*$ its sheaf of invertible elements, and $\mathcal{O}_X^* \subseteq \mathcal{K}^*$ the sheaf of units of $\mathcal{O}_X$. The quotient sheaf $\mathcal{K}^*/\mathcal{O}_X^*$ is the sheafification of $U \mapsto \mathcal{K}^*(U)/\mathcal{O}_X^*(U)$.

Definition 33 (Cartier divisor)

A **Cartier divisor** on X is a global section of the sheaf $\mathcal{K}^*/\mathcal{O}_X^*$. Concretely, it is given by a family $\{(U_i, f_i)\}_i$, where $\{U_i\}$ is an open cover of X and $f_i \in K(X)^*$, such that

$$f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j) \quad \text{for all } i, j.$$

It is **principal** if it can be given by a single pair (X, f) with $f \in K(X)^*$.

- So a Cartier divisor is a divisor that is *locally* the divisor of a rational function; the f_i are its **local equations**.
- Cartier divisors can be defined on any scheme and are directly related to line bundles.

Weil Divisors versus Cartier Divisors

Theorem 34

Let X be an integral, separated, noetherian scheme all of whose local rings are unique factorization domains. Then the group of Weil divisors is isomorphic to the group of Cartier divisors,

$$\mathrm{Div}(X) \cong \mathrm{CaDiv}(X) := \Gamma(X, \mathcal{K}^*/\mathcal{O}_X^*),$$

and principal Weil divisors correspond to principal Cartier divisors.

- This applies to a curve X: every $\mathcal{O}_{X,P}$ is a discrete valuation ring, hence a unique factorization domain. The correspondence is explicit:

$$\{(U_i, f_i)\}_i \longmapsto \sum_P v_P(f_i)\, P \quad \text{(for any } i \text{ with } P \in U_i\text{)}.$$

  This is well defined: if $P \in U_i \cap U_j$, then f_i/f_j is a unit at P, so $v_P(f_i) = v_P(f_j)$.
- Consequently $\mathrm{Cl}(X) \cong \mathrm{CaCl}(X) := \mathrm{CaDiv}(X)/\{\text{principal Cartier divisors}\}$.

Divisors and Line Bundles

- To a divisor D with local equations $\{(U_i, f_i)\}$ we associate the subsheaf $\mathcal{O}_X(D) \subseteq \mathcal{K}$ generated by f_i^{-1} on U_i. It is locally isomorphic to $\mathcal{O}_X$, i.e. an **invertible sheaf** (line bundle).
- $D \sim D'$ if and only if $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$, and $D \mapsto \mathcal{O}_X(D)$ induces an isomorphism

$$\mathrm{Cl}(X) \cong \mathrm{CaCl}(X) \cong \mathrm{Pic}(X),$$

  where $\mathrm{Pic}(X)$ is the group of invertible sheaves up to isomorphism, under $\otimes$.
- Global sections answer the question from the motivation:

$$L(D) := \Gamma(X, \mathcal{O}_X(D)) = \{f \in K(X)^* \mid \mathrm{div}(f) + D \geq 0\} \cup \{0\}, \qquad \ell(D) := \dim_k L(D).$$

The Riemann–Roch Theorem

Theorem 35 (Riemann–Roch)

Let X be a curve of genus g with canonical divisor K_X. For every divisor D on X,

$$\ell(D) - \ell(K_X - D) = \deg D + 1 - g.$$

Example 36

On $\mathbb{P}^1_k$ we have $g = 0$. For $n \geq 0$, $L(n[\infty])$ is the space of polynomials in x of degree at most n, so $\ell(n[\infty]) = n + 1 = \deg(n[\infty]) + 1 - g$.

- The theorem relates the number of linearly independent rational functions with prescribed poles to two numbers: the degree of D and the genus of X.
- Its proof, as a corollary of the Grothendieck–Riemann–Roch theorem, is the goal of the next seminar.

Conclusion

- The Nullstellensatz translates the geometry of algebraic sets in k^n into the algebra of ideals of $k[X_1, \ldots, X_n]$. This motivates $\operatorname{Spec} A$, whose points are the prime ideals of an arbitrary ring A.
- Sheaves describe data that are defined locally and can be glued. A scheme is a locally ringed space that is locally an affine scheme $(\operatorname{Spec} A, \mathcal{O})$.
- On a curve X, every closed point P gives a valuation v_P on the function field. A Weil divisor is a formal sum of closed points, and $\operatorname{div}(f)$ records the zeros and poles of f.
- A Cartier divisor is given locally by one rational function. On a curve, Weil and Cartier divisors agree, and

$$\operatorname{Cl}(X) \cong \operatorname{CaCl}(X) \cong \operatorname{Pic}(X).$$

- The space $L(D)$ of functions with poles bounded by D is the space of global sections of the line bundle $\mathcal{O}_X(D)$; its dimension is governed by the Riemann–Roch theorem.

Next presentation's plan

1. To apply scheme theory to divisors of algebraic curves in more depth (invertible sheaves, the canonical divisor, cohomology of sheaves).
2. To prove the Grothendieck–Riemann–Roch (GRR) theorem using scheme theory.
3. To prove the Riemann–Roch (RR) theorem for curves as a corollary of the GRR theorem.

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Thank you